Aryan Bansal

Phone: 8368857069 | E-mail: aryan.bansal.ug24@nsut.ac.in | CGPA: 7.8 |
LinkedIn: Aryan Bansal | LinkedIn | GitHub - AryanBansal2421

EDUCATION

B.Tech – Instrumentation & Control Engineering (ICE)	2024-2028	Netaji Subhas University of Technology (NSUT)	CGPA – 7.8
CBSE (Class XII)	2024	SD Public School	86.8%
CBSE (Class X)	2022	Venkateshwar Global School	94.8%

EXPERIENCE

1. **Industrial Training, R&D Intern, Wheel Speed Sensor Dept.**
   Continental Automotive India Pvt. Ltd., IMT Manesar
   GitHub - AryanBansal2421/Wheel_Speed_Sensor-
   - Built an automated LabVIEW test bench using FFT-based frequency analysis and a dual-threshold (frequency + amplitude) comparator to validate WSS analog output against spec, eliminating manual oscilloscope debug.
   - Worked directly with raw analog sensor signals — captured and digitized WSS amplitude/frequency output to design the digital pass/fail logic behind the test bench.
   - Ran functional testing on 500+ WSS units/day on PLC-integrated lines, maintaining signal-integrity checks at production throughput.

PROJECTS

1. **Digital Clock Frequency Monitor for SoC Verification | SystemVerilog**
   GitHub – AryanBansal2421/Digital-Clock-Frequency-Monitor-
   - Designed an RTL-based clock frequency monitoring module in System Verilog using counters and registers for clock measurement and validation.
   - Implemented logic to count clock cycles, calculate frequency, and detect frequency deviations from the expected range.
   - Created a self-checking verification environment to test multiple clock conditions and validate correct pass/fail detection through simulation.

2. **SPI Master Controller with Verification Environment | SystemVerilog, RTL Design**
   GitHub – AryanBansal2421/SPI-Master-Controller-with-Verification-Environment
   - Designed an FSM-based SPI Master Controller in SystemVerilog to enable synchronous serial data transmission and reception with external slave devices.
   - Implemented clock generation, chip-select control, and shift-register based data transfer logic to manage SPI communication sequences.
   - Developed a simulation-based verification environment to test different data patterns, validate protocol timing, and analyze signal waveforms for correct operation.

3. **8-bit SAR ADC Design and Performance Analysis | LTspice, MATLAB, Verilog**
   GitHub: AryanBansal2421/8-bit-SAR-ADC-Design-and-Performance-Analysis
   - Designed and simulated a comparator circuit in LTspice to analyze switching behavior and decision thresholds for ADC applications.
   - Implemented an 8-bit R-2R ladder DAC in LTspice and evaluated its digital-to-analog transfer characteristics.
   - Developed a behavioral SAR conversion model using a binary-search algorithm to generate digital output codes for analog input voltages.
   - Characterized comparator and DAC performance through simulation and studied their impact on SAR ADC operation.
   - Analyzed ADC metrics including resolution, LSB size, quantization error, and conversion time.

TECHNICAL SKILLS & INTERESTS

- **Languages:** C, C++, Embedded C, Verilog HDL, Assembly programming
- **Tools & Technologies:** LTspice, MATLAB, LabVIEW, Arduino IDE, Tinkercad Circuits, Oscilloscope & Bench Debug, Git, GitHub
- **Areas of Interest:** Digital Design, Analog & Mixed-Signal Electronics, Data Converters, Embedded Systems, Signal Processing, Microprocessors & Microcontrollers, STA, FPGA (basics).
- **Relevant coursework:** Digital Circuits & Systems, Electronic Devices & Circuits, Analog Electronics, Circuit Analysis, Signals & Systems, Control Systems, Electronic Instrumentation, Power Apparatus, Microprocessor Based System Design.

ACHIEVEMENTS

- Texas Instruments BYTE'26 Mentee
- Smart India Hackathon 2025 Finalists